SMART DOCTOR APPOINTMENT SCHEDULER
Using Forward Chaining (Rule-Based AI)

Submitted by
Name: Kartik Mathur
Registration Number: 25BCE11381
Course: CSA2001 – Fundamentals of AI and ML
Project Type: Bring Your Own Project (BYOP)
Date: March 2026

TITLE

Smart Doctor Appointment Scheduler using Forward Chaining (Rule-Based Artificial Intelligence)

INTRODUCTION

This project develops a Command Line Interface (CLI) based intelligent system that schedules doctor appointments using the Forward Chaining technique of Artificial Intelligence.

MOTIVATION OF THE PROJECT

In college hospitals and clinics, patients often face long waiting times and difficulty in finding available doctors according to their speciality and preferred time. This project

aims to solve this real-world problem using AI concepts learned in the course.

PROBLEM STATEMENT

To create an intelligent rule-based system that automatically matches patient requirements (speciality and time) with available doctors using Forward Chaining and provides instant confirmation or alternative suggestions.

OBJECTIVE OF THE PROJECT

- Implement Forward Chaining for intelligent decision making
- Provide a simple and user-friendly CLI interface
- Demonstrate practical application of AI concepts

EXISTING METHODS

Traditional appointment systems rely on manual booking or simple web forms without applying any AI-based reasoning.

PROS AND CONS OF EXISTING METHODS

Pros:
Simple to understand and use

Cons:
No intelligent rule checking

Long waiting time
Possibility of human error

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware: Any computer system
Software: Python 3.x

METHODOLOGY AND GOAL

The system uses production rules and Forward Chaining to infer the best possible appointment. Patient input is taken as facts, and rules are applied sequentially to reach a decision.

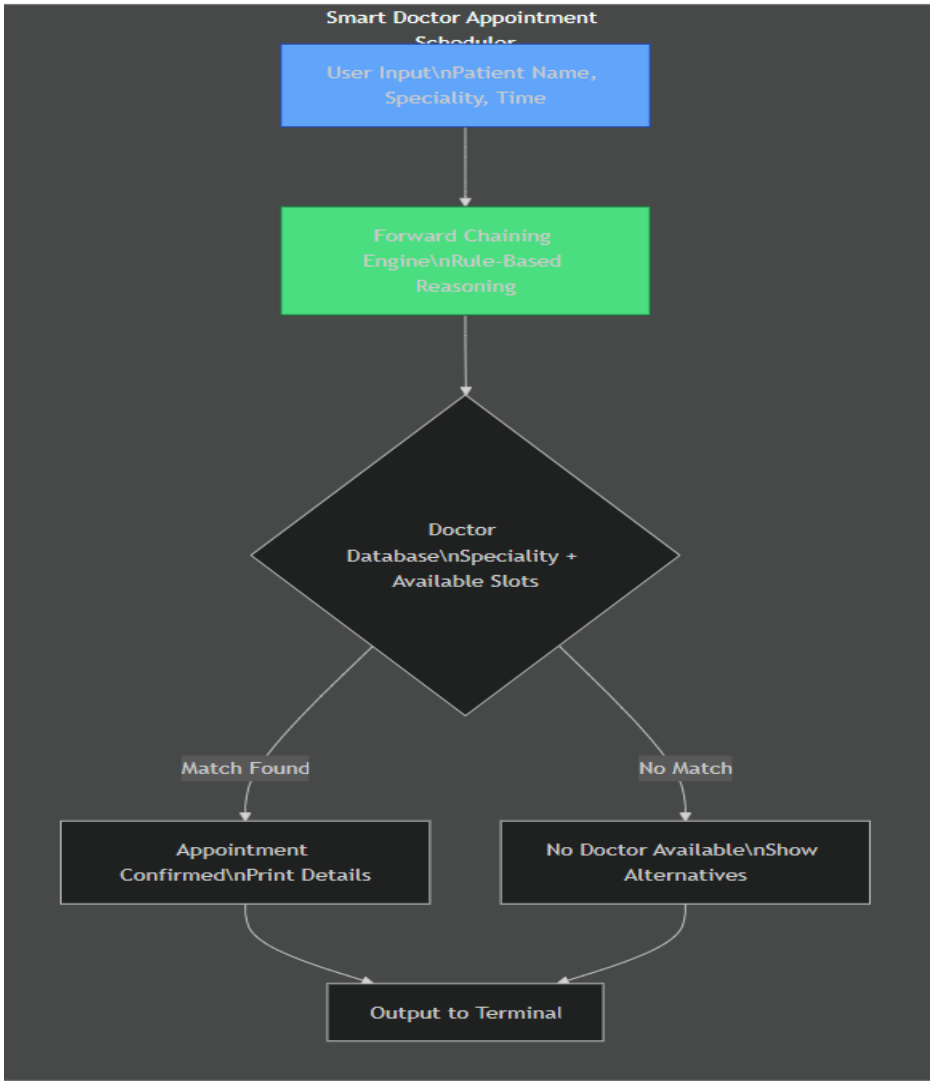
FUNCTIONAL MODULES DESIGN AND ANALYSIS

1. Doctor Database Module
2. User Input Module
3. Forward Chaining Reasoning Module
4. Output and Confirmation Module

ALGORITHM DEVELOPMENT

Forward Chaining Algorithm:
Start with known facts (patient requirements)
Apply rules one by one
Derive a conclusion (appointment booked or not available)

SOFTWARE DIAGRAM



CODING

The complete implementation is written in Python and available in the file main.py.

OUTPUT

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

> python -u "C:\Users\Asus\AppData\Local\Temp\temp

=== Smart Doctor Appointment Scheduler (Forward Chaining AI) ===
This system uses Forward Chaining to book appointments intelligent

Enter Patient Name: kartik
Enter Speciality (General / Cardiology): General
Enter Preferred Time (e.g. 09:00): 09:00

✅ APPOINTMENT SUCCESSFULLY BOOKED!
Doctor     : Dr. Sharma
Speciality : General
Time       : 09:00
Patient    : kartik

Thank you for using the AI Scheduler!
PS C:\Users\Asus>
```

KEY IMPLEMENTATIONS

- Rule-based Forward Chaining logic
- Multiple doctor support with different specialities
- Real-time slot availability checking

SIGNIFICANT PROJECT OUTCOMES

Successfully developed a working AI-based system using Forward Chaining concepts.

TESTING AND REFINEMENT

The system was tested with multiple inputs including valid and invalid cases. Error messages were refined to improve user experience.

REAL-WORLD APPLICATIONS

This system can be extended to hospital management systems, online booking platforms, and scheduling applications.

CONTRIBUTION / FINDINGS

This project demonstrates how simple AI techniques like Forward Chaining can solve real-life problems efficiently.

LIMITATIONS

- Uses static data
- No graphical interface (CLI-based system)

CONCLUSION

This project successfully applies Artificial Intelligence concepts to solve a real-world scheduling problem.

FUTURE ENHANCEMENTS

- Add database support (SQLite)
- Implement Backward Chaining

- Develop a GUI or web-based interface
- Integrate with real hospital systems

REFERENCES

Course Material – Problem Solving Methods and Knowledge Representation
Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvi